Operating and Maintenance Manual

All Alvery pin modules · Edition 2025-03

1. About this manual

This manual applies to all Alvery pin modules of the METRO, PRO, MICRO and AUTO lines. Module-specific values (dimensions, forces, torques) are given in the data sheets. If a data sheet and this manual differ, the data sheet applies.

Keep this manual near the machine. Read it before installing a module for the first time.

2. Safety

Pin modules are not lifting devices. Never use them to lift loads or persons.

Wear safety glasses when cleaning with compressed air. Pins can be ejected if the pin field cover is removed.

Machine the workpiece only when the pin field is locked. An unlocked pin field has no defined holding force.

Pneumatic and electric modules must be disconnected from the energy supply before maintenance.

3. Installation

PRO modules are mounted in a machine vise via adapter plates or on base plates with connection pins (M10 x 27 mm or M10 x 42 mm). Tighten connection pins with 15 Nm.

METRO modules are mounted on 50 x 100 mm base plates that are fastened to T-slot plates with M8 slot nuts.

AUTO modules are fastened with 4 x M5 screws (tightening torque 6 Nm). Connect pneumatic modules with QS-6 (AutoForm 67, PickForm, FormBed p) or QS-4 hoses (AutoForm Mini, AutoForm Mini Duo).

The FormBed e is connected with an M12 connector (5-pin). Pin 1: +24 V DC, pin 3: 0 V, pin 2: locking signal, pin 4: status output 'locked'.

4. Operation

1. Release the pin field (open clamping screw, open hand lever, vent the pneumatic actuator or send the release signal).

2. Reset the pins to the initial position by turning the module or with the reset plate.

3. Press the workpiece into the pin field until it rests on the desired contour.

4. Lock the pin field. For manual PRO modules always use the torque wrench; the holding force depends directly on the tightening torque.

5. Check that no pin is at its end position. If outer pins are fully pushed in, use a module with a larger stroke or a height adapter.

5. Tightening torques and holding forces (PRO line)

Module	Order no.	Rated torque	Holding force at rated torque
FormVise 100 (symmetric)	41.0672	25 Nm	7,5 kN
FormVise 125 (symmetric)	41.0852	25 Nm	9,0 kN
FormVise 127W (symmetric)	41.1272	30 Nm	9,5 kN
FormVise 140 H (high, symmetric)	41.0674	25 Nm	7,5 kN
FormVise 140 L (low, symmetric)	41.0675	25 Nm	7,5 kN

Module	Order no.	Rated torque	Holding force at rated torque
FormVise 40 Mini	41.0386	12 Nm	1,8 kN
FormRest 50	41.0312	25 Nm	3,7 kN

Exceeding the rated torque does not increase the holding force and damages the thread of the clamping screw. For FormVise 125 units affected by a service notice, follow the torque given in the notice.

6. Cleaning and maintenance

Line	Cleaning	Lubrication	Inspection / wear
PRO	Blow out the pin field with compressed air after every shift.	Lubricate the pins with Alvery pin oil (49.9100) every 250 operating hours.	No wear-relevant parts apart from pins and clamping screw.
METRO	Wipe the housing and pin tips with a lint-free cloth once a week.	No lubrication required. Do not oil the pins (dust accumulation).	Inspect the hand lever mechanism once a year.
MICRO	Clean with compressed air at max. 2 bar. Ultrasonic cleaning is not permitted.	No lubrication required.	Bent pins must be replaced; straightening is not permitted.
AUTO	Blow out the pin field with compressed air every 8 hours of operation.	Pneumatic actuators are lubricated for life. Pins: Alvery pin oil (49.9100) every 500 operating hours.	Rated for 2 million clamping cycles; replace the seal kit after 2 million cycles.

Removing the pin field cover: loosen the four M4 screws of the cover, lift it off and remove the pins row by row. Do not mix pins of different modules. Reassemble in reverse order and tighten the cover screws with 2 Nm.

7. Troubleshooting

Problem	Possible cause	Remedy
Pins do not return to the initial position after releasing	Chips or dried coolant in the pin field	Blow out with compressed air; if necessary, remove the pin field cover and clean the pins (see section 6).
Holding force too low	Clamping screw not tightened to rated torque, or worn screw thread	Tighten with the torque wrench to the value in the data sheet. Check the thread; replace the screw if worn.
Workpiece moves during machining	Pin stroke exhausted (pins at end position)	Use a module with a larger pin stroke or add a height adapter so the contour lies within the stroke.
Pneumatic module does not lock	Supply pressure below 5 bar or leaking hose connection	Check the supply pressure (6 bar) and the QS connection; replace the seal kit 44.9067 if the actuator leaks.
FormBed e does not lock	Supply voltage outside 24 V DC ± 10 % or control signal missing	Check supply voltage and the locking signal (pin 2 of the M12 connector).
Artefacts in CT scan	Steel parts in the beam path	Use only carbon pin extensions and position base plates outside the beam.
Single pins stuck	Bent pin	Replace the pin. Do not straighten bent pins.

8. Spare parts

Article no.	Description
41.9006	Replacement pins Ø 6 mm, pack of 10
41.9010	Clamping screw M10, hardened
42.6759	Replacement foam tips, pack of 100
42.9006	Replacement pins Ø 6 mm, pack of 10
42.9020	Hand lever, complete
43.9001	Replacement pins Ø 1 mm, pack of 50
44.9067	Seal kit for pneumatic actuator
45.9014	Replacement pins Ø 14 mm, pack of 10

Article no.	Description
45.9031	Electric locking actuator, 24 V DC
49.9100	Alvery pin oil, 100 ml

9. Storage and disposal

Store modules dry at +5 °C to +40 °C with the pin field released. Oil PRO modules lightly before storage of more than 3 months. Modules consist of aluminium, stainless steel and small amounts of plastics; dispose of them via metal recycling.